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Mini Project Report on

"Medicine Reminder App"

Submitted in partial fulfillment of the academic year 2021-2022

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the mini project on topic "**Medicine Reminder App**" has been successfully submitted by **ABHILASH MADHAV M (1AH19CS002)**, **MOHIT KUMAR M (1AH18CS053)**, **NANDAN R (1AH18CS057)** bonafide students of ACS College of Engineering affiliated to Visveswaraya Technological University, Belagavi during the year 2021-2022. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report submitted to the department. The mini project has been approved as it satisfies the academic requirements in respect of mini project work as prescribed in the 6th semester.

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ABSTRACT

This is an Android-based application in which an automatic alarm ringing system is implemented. It focuses on doctor and patient interaction. Patients need not remember their medicine dosage timings as they can set an alarm on their dosage timings. The alarm can be set for multiple medicines and timings including date, time and medicine description.

A notification will be sent to them through email or message inside the system preferably chosen by the patients. They can search doctor disease wise. The patients will get the contact details of doctors as per their availability. Also the users can see different articles related to medical fields and health care tips. The system focuses on easy navigation and good user interface.

Many such Medical Reminder Systems have been developed where a new hardware is required but in our work we have made an attempt to develop a system which is economical, time-saving and supports medication adherence.

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CHAPTER 1

PREAMBLE

1.1 Introduction

The category of patients involve all human beings-teachers, students, businessmen, housewives, children and also all of us have a busy hectic schedule. Today's life is full of responsibilities and stress. So people are prone to diseases of different types and it is our duty to make ourselves stay fit and healthy. If the patient stays at home then he or she might get someone to look after him/her but when one is not at home, is out of the city or state away from home then it is hard for the family members to call them and remind them their dosage timings every time. In our developing and technology dependent life we totally rely on gadgets especially smart phones. Today everyone has a smart phone. With this we get an opportunity to use technology in a better way so that it can be made useful to us. And it plays an important part in our daily life and helps us staying fit in many ways. The remarkable problem is that patients forget to take the proper medicines in proper proportion and in proper time.

Medication adherence, which refers to the degree or extent to which a patient takes the right medication at the right time according to a doctor's prescription, has recently emerged as a serious issue because many studies have reported that non-adherence may critically affect the patient, thereby raising medical costs. Medication non adherence is a common, complex, and costly problem that contributes to poor treatment outcomes and consumes health care resources

So we are introducing an Android application whose objective is to remind the patients of their dosage timings through Alarm Ringing system so that they can stay fit and healthy. Through navigation they can search doctors and hospitals and contact details so that they can easily get proper treatment on time. This application focuses on the people who forget to take medicines on time. It allows users to set an alarm along with the fields of date, time and medicine description which will allow them to set alarm for multiple medicines at different time intervals. The notification system will send a notification after setting an alarm. The user can activate or deactivate the notification accordingly. It will be sent as email or message as selected by the user. The patients can search doctor disease wise and area wise which will provide easy searching facility along with doctor's contact information, visiting place and availability time. Medication reminders help in decreasing medication dispensing errors and wrong dosages.

1.2 Objective

The application is designed on Android Studio. It can be helpful in defense sector and emergency conditions (accidents) and can spread health care awareness. It is life-saving, money saving and time saving application which is easy to use and provides a good user interface. We are introducing an Android application whose objective is to remind the patients of their dosage timings through Alarm Ringing system so that they can stay fit and healthy. Through navigation they can search doctors and hospitals and contact details so that they can easily get proper treatment on time. This application focusses on the people who forget to take medicines on time. It allows users to set an alarm along with the fields of date, time and medicine description which will allow them to set alarm for multiple medicines at different time intervals. The notification system will send a notification after setting an alarm. The user can activate or deactivate the notification accordingly. It will be sent as email or message as selected by the user. The patients can search doctor disease wise and area wise which will provide easy searching facility along with doctor's contact information, visiting place and availability time. Medication reminders help in decreasing medication dispensing errors and wrong dosages.

1.3 Scope

The users will get the notifications through SMS also. It will provide the information about the medicine timings. The scheduled appointment with the doctor with the contact details including visiting time, venue and availability at different hospitals in case the appointment is missed at the scheduled place. The new appointment will be set accordingly. The system focuses on improving the rate of attendance at healthcare appointments. The personal phone notifications and reminders are a strong supporting tool in improving medication adherence strategies. The New England Healthcare Institute estimates that \$290 billion of healthcare expenditures could be avoided if medication adherence were improved. It supports an easy implementation as it is less expensive, reliable, scalable, accessible to anyone with smartphones, and do not require separate devices, packaging or extra hardware. In case if the users phone is switched off and he has set the alarm and the notification is set on , still he will be able to get the notifications through email or message(on his device as well as on other registered number), so it works even when you are running out of the battery. Also a facility of reminding the doctor's next appointment in the system has been focused. We have also implemented a navigation system which will allow users to locate the nearest registered hospitals according to their current location. The location based searching of the doctors as well as disease wise searching has been focused which makes the application more suitable, more user friendly with great features and satisfactory results.

CHAPTER 2

TECHNOLOGY AND SYSTEM REQUIREMENTS

2.1 Technology Used

Android Studio is the official integrated development environment (IDE) for Google's Android operating system, designed specifically for Android development. Android studio is used to complete this application with the help of following things:

XML

XML tags define the data and used to store and organize data and to design the layouts. It's easily scalable and simple to develop. In Android, the XML is used to implement DI-related data, and only contains tags, while implementing they need to be just invoked.

JAVA

Java is used for coding. This page control all the elements of xml with time. User can give default attributes values for different elements in xml, which will be used in the Activity(app) until the user change that attribute in corresponding JAVA file for that particular element.

2.2 System Requirements

- 64-bit Microsoft Windows 8/10.
- x86_64 CPU architecture, 2nd generation Intel Core or newer, or AMD CPU with support for a Windows Hypervisor.
- 8 GB RAM or more.
- 8 GB of available disk space minimum (IDE+ Android SDK + Android Emulator)
- 1280 x 800 minimum screen resolution.

CHAPTER 3

IMPLEMENTATION

In this project, Medication reminders help in decreasing medication dispensing errors and wrong dosages. The Reminder system consists of two parts –setting Alarm and getting notification.

3.1 Implementation Steps

Step 1: Create a New Project

First create an empty activity. We are going to call this app Simple Weather. Enter a unique package name, set the minimum required SDK to Android 2.2, and set the target **SDK** to Android 4.4.

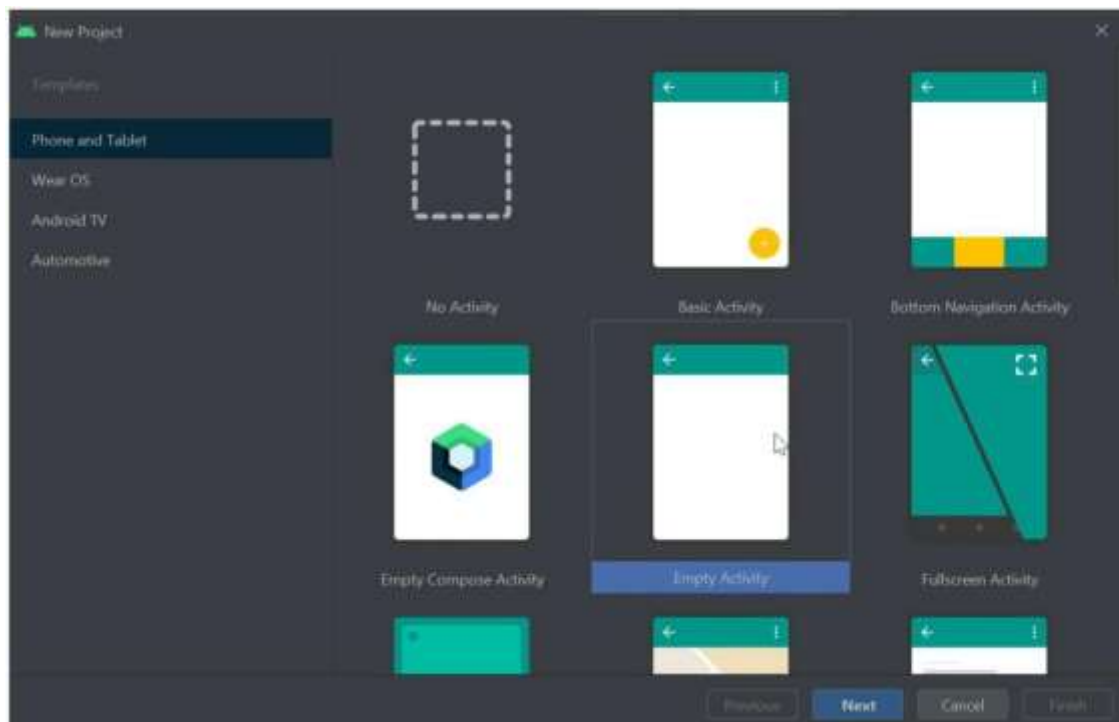
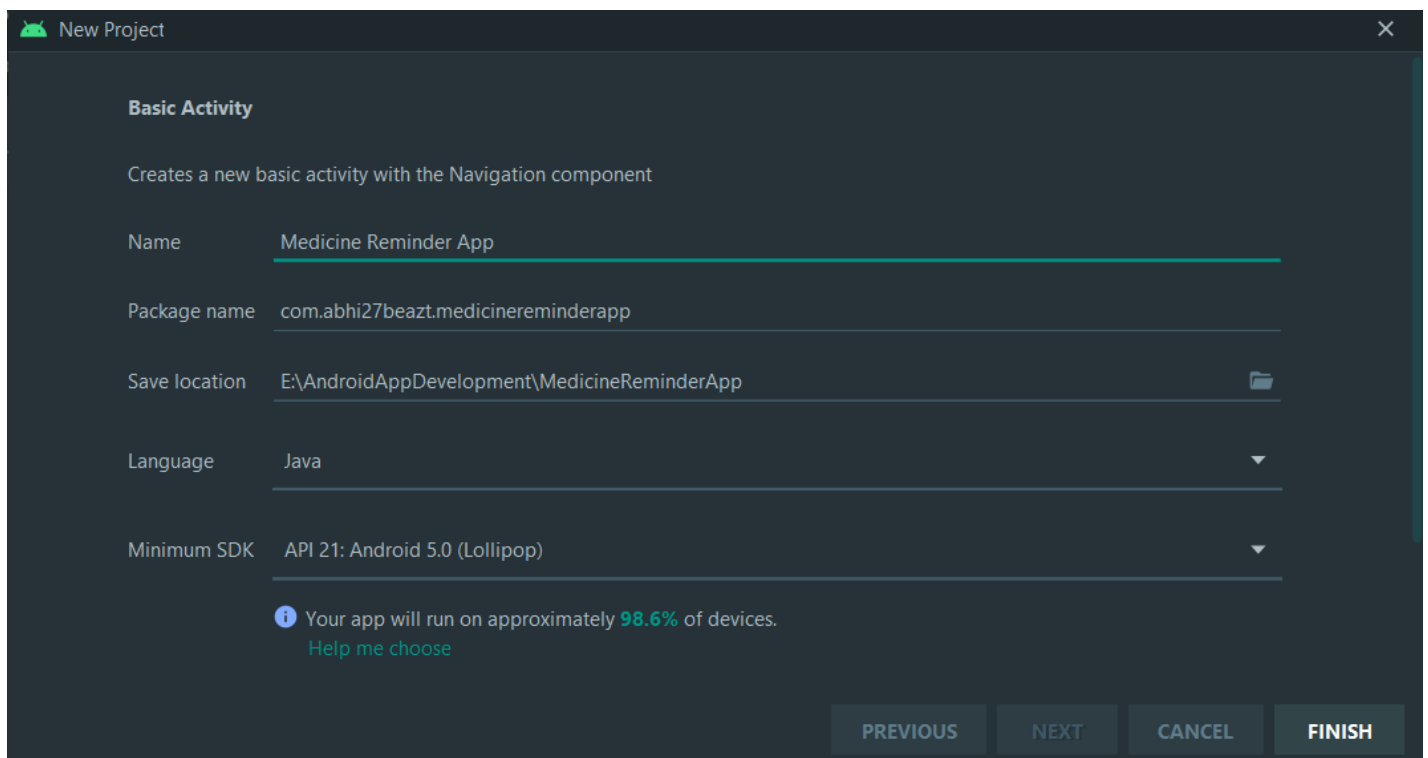


Fig 3.1.1: Creating a New Project



3.1.2: Enter App Details

Step 2: Edit the Manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.abhilash.medicinetime">

    <uses-permission android:name="android.permission.VIBRATE" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/AppTheme.NoActionBar">
        <activity android:name="com.abhilash.medicinetime.medicine.MedicineActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
```

```
<intent-filter>
<action android:name="android.intent.action.MAIN" />
<intent-filter>
    <action android:name="android.intent.action.MAIN" />

    <category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
<!-- <activity android:name=".MainActivity" /> -->
<activity android:name="com.abhilash.medicinetime.addmedicine.AddMedicineActivity" />
<activity android:name="com.abhilash.medicinetime.alarm.ReminderActivity" />
<activity android:name="com.abhilash.medicinetime.report.MonthlyReportActivity"/>
</application>
```

Step 3: Create Medicine App Java File (MedicineApp.java)

```
import android.app.Application;
import android.content.Context;

/**
 * Created by Abhilash on 7/07/22.
 */

public class MedicineApp extends Application {

    private static Context mInstance;

    @Override
    public void onCreate() {
        super.onCreate();
        if (mInstance == null) {
            mInstance = getApplicationContext();
        }
    }

    public static Context getInstance() {
        return mInstance;
    }
}
```

Step 4: Create Medicine Activities (addmedicine)

A) AddMedicineActivity.java

```
package com.abhilash.medicinetime.addmedicine;
```

```
import android.os.PersistableBundle;
import androidx.appcompat.app.ActionBar;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import androidx.appcompat.widget.Toolbar;

import com.abhilash.medicinetime.Injection;
import com.abhilash.medicinetime.R;
import com.abhilash.medicinetime.utils.ActivityUtils;

public class AddMedicineActivity extends AppCompatActivity {

    public static final int REQUEST_ADD_TASK = 1;

    public static final String SHOULD_LOAD_DATA_FROM_REPO_KEY =
"SHOULD_LOAD_DATA_FROM_REPO_KEY";

    public static final String EXTRA_TASK_ID = "task_extra_id";
    public static final String EXTRA_TASK_NAME = "task_extra_name";

    private AddMedicinePresenter mAddMedicinePresenter;

    private ActionBar mActionBar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_add_medicine);

        //Setup toolbar
        Toolbar toolbar = (Toolbar) findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
        mActionBar = getSupportActionBar();
        mActionBar.setDisplayHomeAsUpEnabled(true);
        mActionBar.setDisplayShowHomeEnabled(true);

        AddMedicineFragment addMedicineFragment = (AddMedicineFragment)
getSupportFragmentManager().findFragmentById(R.id.contentFrame);

        int medId = getIntent().getIntExtra(AddMedicineFragment.ARGUMENT_EDIT_MEDICINE_ID,0);
        String medName =
getIntent().getStringExtra(AddMedicineFragment.ARGUMENT_EDIT_MEDICINE_NAME);

        setToolbarTitle(medName);

        if (addMedicineFragment == null) {
            addMedicineFragment = AddMedicineFragment.newInstance();

            if (getIntent().hasExtra(AddMedicineFragment.ARGUMENT_EDIT_MEDICINE_ID)) {
                Bundle bundle = new Bundle();
```

```

bundle.putInt(AddMedicineFragment.ARGUMENT_EDIT_MEDICINE_ID, medId);
    addMedicineFragment.setArguments(bundle);
}
    ActivityUtils.addFragmentToActivity(getSupportFragmentManager(),
        addMedicineFragment, R.id.contentFrame);
}

    boolean shouldLoadDataFromRepo = true;
    // Prevent the presenter from loading data from the repository if this is a config change.
    if (savedInstanceState != null) {
        // Data might not have loaded when the config change happen, so we saved the state.
        shouldLoadDataFromRepo =
savedInstanceState.getBoolean(SHOULD_LOAD_DATA_FROM_REPO_KEY);
    }

//    // Create the presenter
    mAddMedicinePresenter = new AddMedicinePresenter(
        medId,
        Injection.provideMedicineRepository(getApplicationContext()),
        addMedicineFragment,
        shouldLoadDataFromRepo);

}

public void setToolbarTitle(String medicineName) {
    if (medicineName == null) {
        mActionBar.setTitle(getString(R.string.new_medicine));
    } else {
        mActionBar.setTitle(medicineName);
    }
}

@Override
public void onSaveInstanceState(Bundle outState, PersistableBundle outPersistentState) {
    outState.putBoolean(SHOULD_LOAD_DATA_FROM_REPO_KEY,
mAddMedicinePresenter.isDataMissing());
    super.onSaveInstanceState(outState, outPersistentState);
}

@Override
public boolean onSupportNavigateUp() {
    onBackPressed();
    return true;
}
}

```

B) AddMedicineContract.java

```

package com.abhilash.medicinetime.addmedicine;

import com.abhilash.medicinetime.BasePresenter;
import com.abhilash.medicinetime.BaseView;

```

```
import com.abhilash.medicinetime.data.source.MedicineAlarm;
import com.abhilash.medicinetime.data.source.Pills;

import java.util.List;

/**
 * Created by gautam on 12/07/17.
 */

public interface AddMedicineContract {

    interface View extends BaseView<Presenter> {

        void showEmptyMedicineError();

        void showMedicineList();

        boolean isActive();

    }

    interface Presenter extends BasePresenter{

        void saveMedicine(MedicineAlarm alarm, Pills pills);

        boolean isDataMissing();

        boolean isMedicineExists(String pillName);

        long addPills(Pills pills);

        Pills getPillsByName(String pillName);

        List<MedicineAlarm> getMedicineByPillName(String pillName);

        List<Long> tempIds();

        void deleteMedicineAlarm(long alarmId);

    }

}
```

C) AddMedicinePresenter.java

```
package com.abhilash.medicinetime.addmedicine;

import androidx.annotation.NonNull;

import com.abhilash.medicinetime.data.source.MedicineAlarm;
```

```
import com.abhilash.medicinetime.data.source.MedicineDataSource;
import com.abhilash.medicinetime.data.source.Pills;

import java.util.List;

/**
 * Created by gautam on 12/07/17.
 */

public class AddMedicinePresenter implements AddMedicineContract.Presenter,
MedicineDataSource.GetTaskCallback {

    @NonNull
    private final MedicineDataSource mMedicineRepository;

    private final AddMedicineContract.View mAddMedicineView;

    private int mMedicineId;

    private boolean mIsDataMissing;

    public AddMedicinePresenter(int mMedicineId, @NonNull MedicineDataSource mMedicineRepository,
AddMedicineContract.View mAddMedicineView, boolean mIsDataMissing) {
        this.mMedicineId = mMedicineId;
        this.mMedicineRepository = mMedicineRepository;
        this.mAddMedicineView = mAddMedicineView;
        this.mIsDataMissing = mIsDataMissing;

        mAddMedicineView.setPresenter(this);
    }

    private boolean isNewTask() {
        return mMedicineId <= 0;
    }

    @Override
    public void start() {

    }

    @Override
    public void saveMedicine(MedicineAlarm alarm, Pills pills) {
        mMedicineRepository.saveMedicine(alarm, pills);
    }

    @Override
    public boolean isDataMissing() {
        return mIsDataMissing;
    }

    @Override
    public boolean isMedicineExists(String pillName) {
```

```
return mMedicineRepository.medicineExists(pillName);
}

@Override
public long addPills(Pills pills) {
    return mMedicineRepository.savePills(pills);
}

@Override
public Pills getPillsByName(String pillName) {
    return mMedicineRepository.getPillsByName(pillName);
}

@Override
public List<MedicineAlarm> getMedicineByPillName(String pillName) {
    return mMedicineRepository.getMedicineByPillName(pillName);
}

@Override
public List<Long> tempIds() {
    return mMedicineRepository.tempIds();
}

@Override
public void deleteMedicineAlarm(long alarmId) {
    mMedicineRepository.deleteAlarm(alarmId);
}

@Override
public void onTaskLoaded(MedicineAlarm medicineAlarm) {
    // The view may not be able to handle UI updates anymore
    /*if (mAddMedicineView.isActive()){
        mAddMedicineView.setDose(medicineAlarm.getDose());
        mAddMedicineView.setMedName(medicineAlarm.getMedicineName());
        mAddMedicineView.setDays(medicineAlarm.getDays());
        mAddMedicineView.setTime(medicineAlarm.getTime());
    }
    mIsDataMissing = false;*/
}

@Override
public void onDataNotAvailable() {
    if (mAddMedicineView.isActive()) {
        mAddMedicineView.showEmptyMedicineError();
    }
}
}
```

Step 5: Creating Alarm Activities

A) ReminderActivity.java

```
package com.abhilash.medicinetime.alarm;

import android.content.Intent;
import android.os.Bundle;

import androidx.appcompat.app.ActionBar;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import android.view.MenuItem;

import com.abhilash.medicinetime.Injection;
import com.abhilash.medicinetime.R;
import com.abhilash.medicinetime.utils.ActivityUtils;

import butterknife.BindView;
import butterknife.ButterKnife;

public class ReminderActivity extends AppCompatActivity {

    @BindView(R.id.toolbar)
    Toolbar toolbar;
    ReminderPresenter mReminderPresenter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_reminder_activity);
        ButterKnife.bind(this);
        setSupportActionBar(toolbar);
        ActionBar actionBar = getSupportActionBar();
        if (actionBar != null) {
            actionBar.setDisplayHomeAsUpEnabled(true);
        }

        Intent intent = getIntent();
        if (!intent.hasExtra(ReminderFragment.EXTRA_ID)) {
            finish();
            return;
        }
        long id = intent.getLongExtra(ReminderFragment.EXTRA_ID, 0);
        ReminderFragment reminderFragment = (ReminderFragment)
        getSupportFragmentManager().findFragmentById(R.id.contentFrame);
        if (reminderFragment == null) {
            reminderFragment = ReminderFragment.newInstance(id);
            ActivityUtils.addFragmentToActivity(getSupportFragmentManager(), reminderFragment,
            R.id.contentFrame);
        }
    }
}
```

```
//Create MedicinePresenter
mReminderPresenter = new
ReminderPresenter(Injection.provideMedicineRepository(ReminderActivity.this), reminderFragment);
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    if (item.getItemId() == android.R.id.home) {
        if (mReminderPresenter != null) {
            mReminderPresenter.finishActivity();
        }
    }
    return super.onOptionsItemSelected(item);
}

@Override
public void onBackPressed() {
    if (mReminderPresenter != null) {
        mReminderPresenter.finishActivity();
    }
}
}
```

B) ReminderContract.java

```
package com.abhilash.medicinetime.alarm;

import com.abhilash.medicinetime.BasePresenter;
import com.abhilash.medicinetime.BaseView;
import com.abhilash.medicinetime.data.source.History;
import com.abhilash.medicinetime.data.source.MedicineAlarm;

/**
 * Created by gautam on 13/07/17.
 */

public interface ReminderContract {

    interface View extends BaseView<Presenter> {

        void showMedicine(MedicineAlarm medicineAlarm);

        void showNoData();

        boolean isActive();

        void onFinish();

    }

    interface Presenter extends BasePresenter {

        void finishActivity();

    }

}
```

```
void onStart(long id);

    void loadMedicineById(long id);

    void addPillsToHistory(History history);

}
}
```

C) ReminderPresenter.java

```
package com.abhilash.medicinetime.alarm;

import androidx.annotation.NonNull;

import com.abhilash.medicinetime.data.source.History;
import com.abhilash.medicinetime.data.source.MedicineAlarm;
import com.abhilash.medicinetime.data.source.MedicineDataSource;
import com.abhilash.medicinetime.data.source.MedicineRepository;

/**
 * Created by gautam on 13/07/17.
 */

public class ReminderPresenter implements ReminderContract.Presenter {

    private final MedicineRepository mMedicineRepository;

    private final ReminderContract.View mReminderView;

    ReminderPresenter(@NonNull MedicineRepository medicineRepository, @NonNull
ReminderContract.View reminderView) {
        this.mMedicineRepository = medicineRepository;
        this.mReminderView = reminderView;

        mReminderView.setPresenter(this);
    }

    @Override
    public void start() {

    }

    @Override
    public void finishActivity() {
        mReminderView.onFinish();
    }

    @Override
    public void onStart(long id) {
        loadMedicineById(id);
    }
}
```

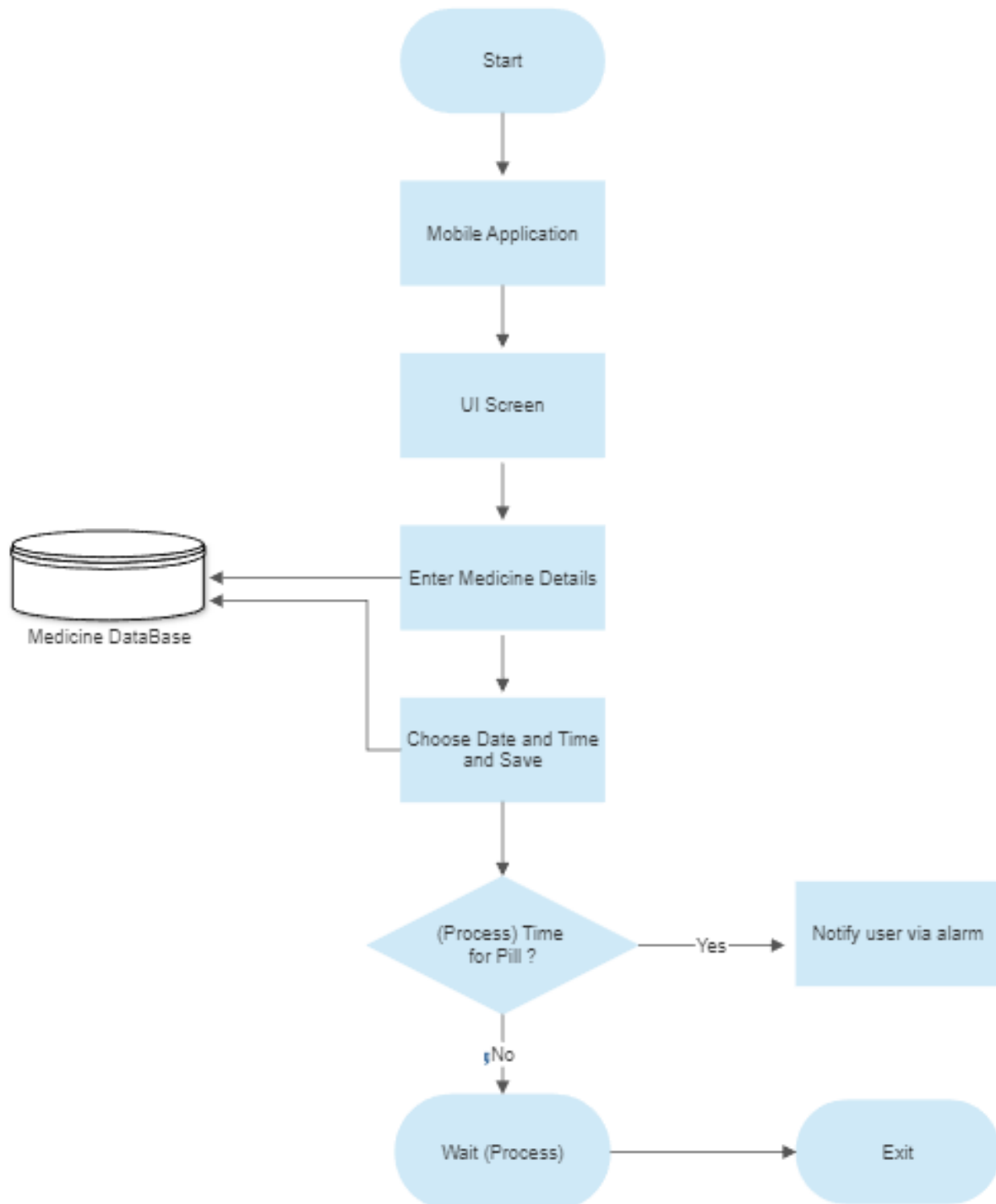
```
@Override
public void loadMedicineById(long id) {
    loadMedicine(id);
}

private void loadMedicine(long id) {
    mMedicineRepository.getMedicineAlarmById(id, new MedicineDataSource.GetTaskCallback() {
        @Override
        public void onTaskLoaded(MedicineAlarm medicineAlarm) {
            if (!mReminderView.isActive()) {
                return;
            }
            if (medicineAlarm == null) {
                return;
            }
            mReminderView.showMedicine(medicineAlarm);
        }
    });

    @Override
    public void onDataNotAvailable() {
        mReminderView.showNoData();
    }
}

@Override
public void addPillsToHistory(History history) {
    mMedicineRepository.saveToHistory(history);
}
}
```

3.2 Flow Chart



CHAPTER 4

RESULTS

4.1 Result snippets

5

Fig 4.1.1: Tap on Add Button

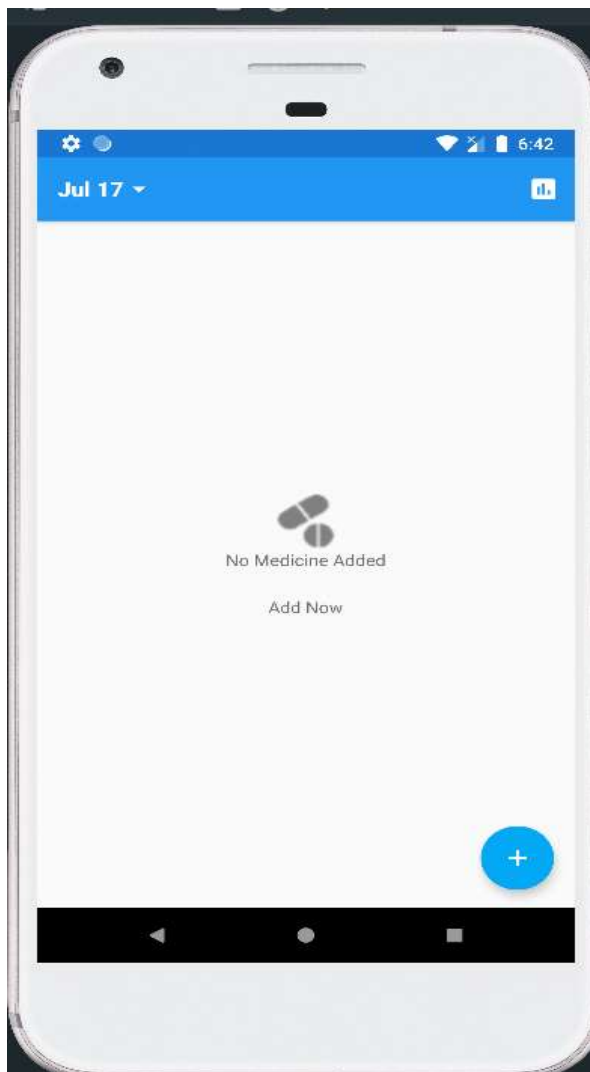


Fig 4.1.2: Choose Date

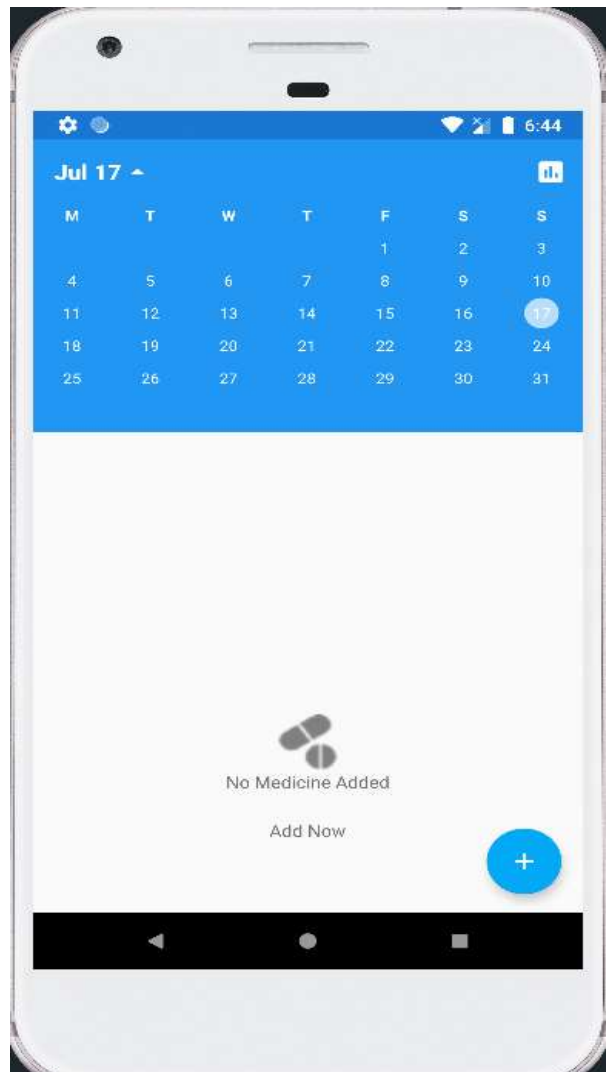


Fig 4.1.3: Enter Time to set alarm



Fig 4.1.4: Add Medicine Details

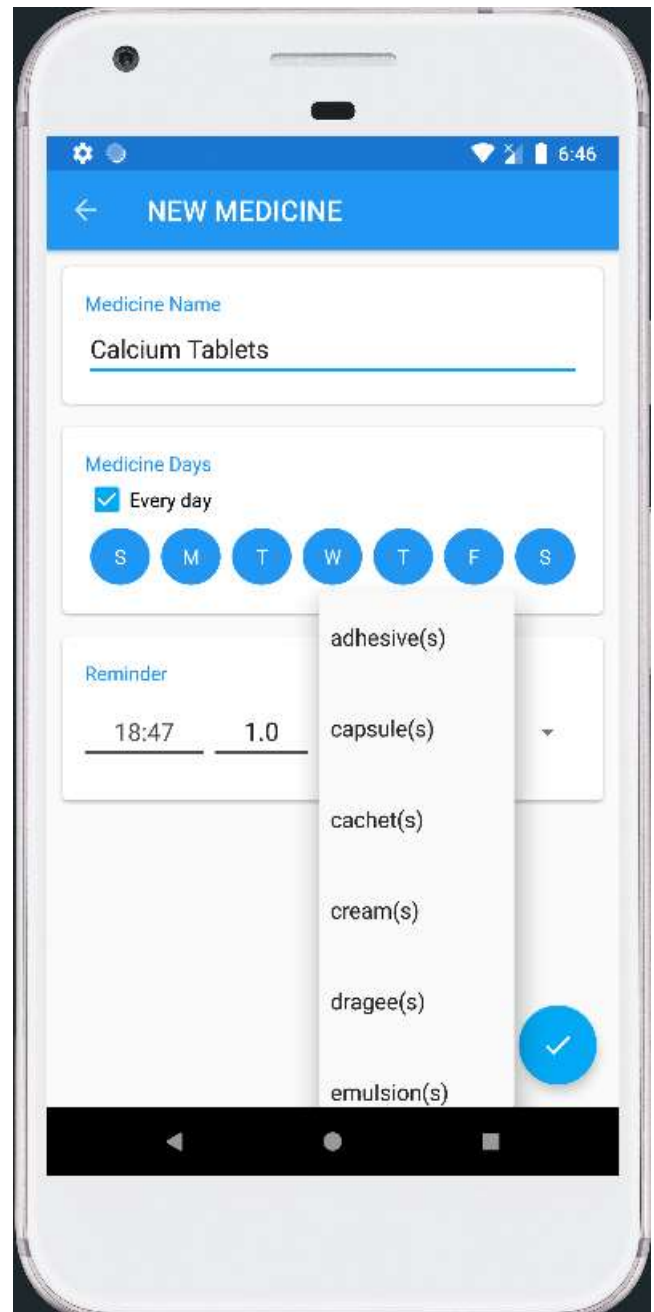


Fig 4.1.5: Medicine Added Successfully

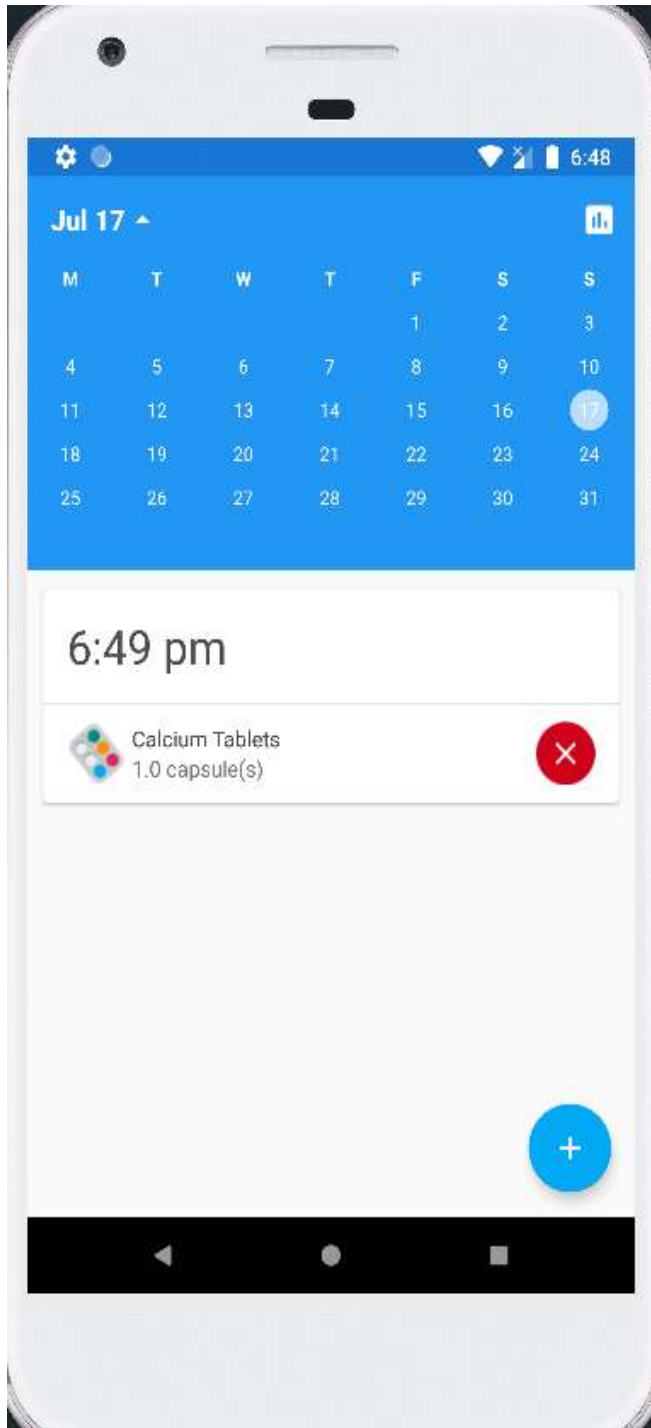


Fig 4.1.6: Track previous pills

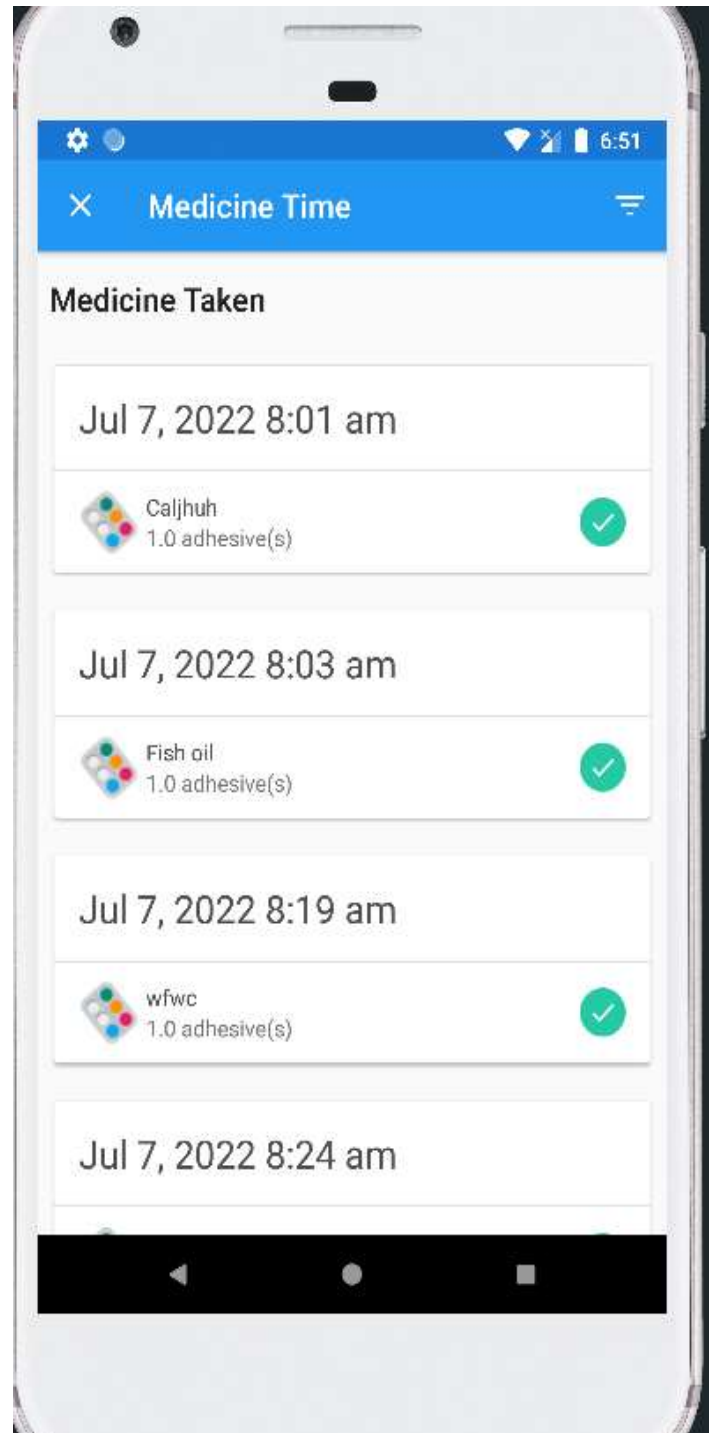


Fig 4.1.7: View taken or ignored pills

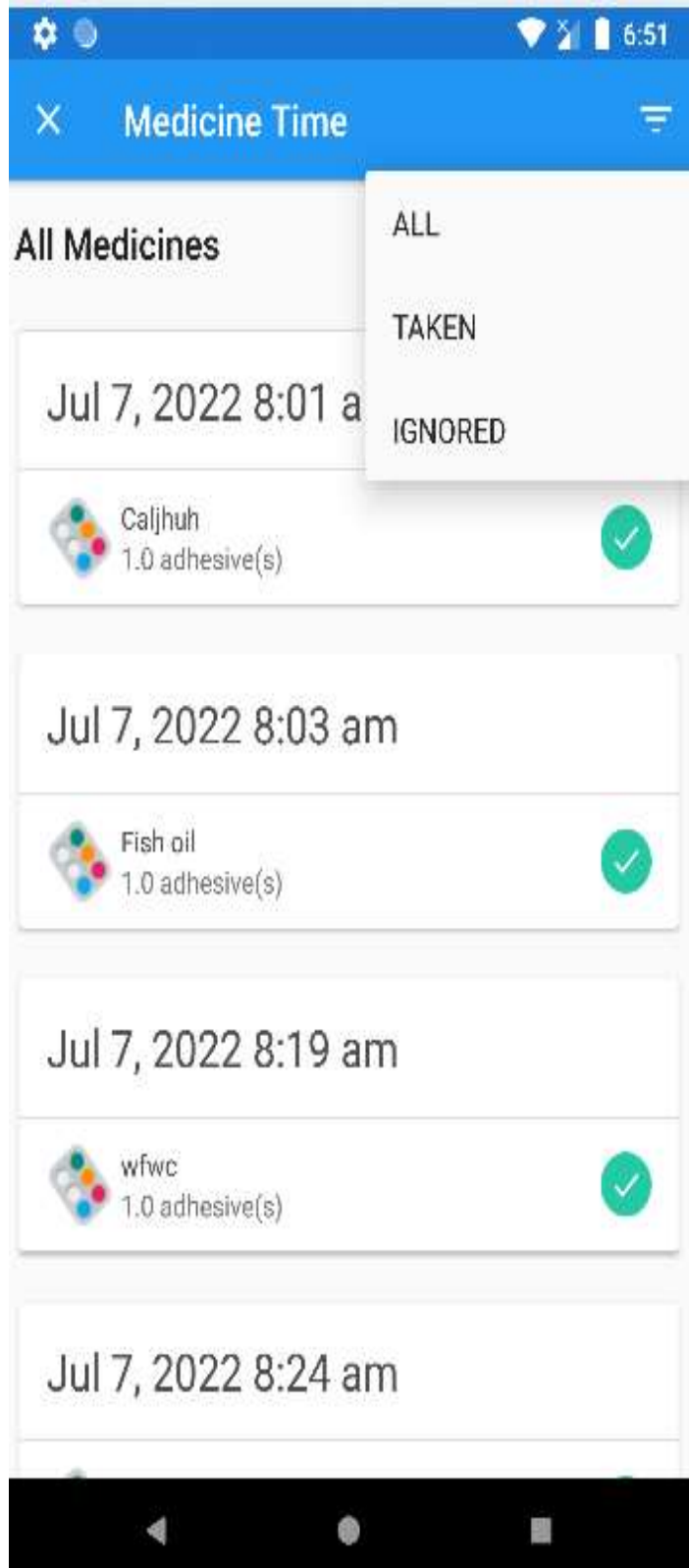
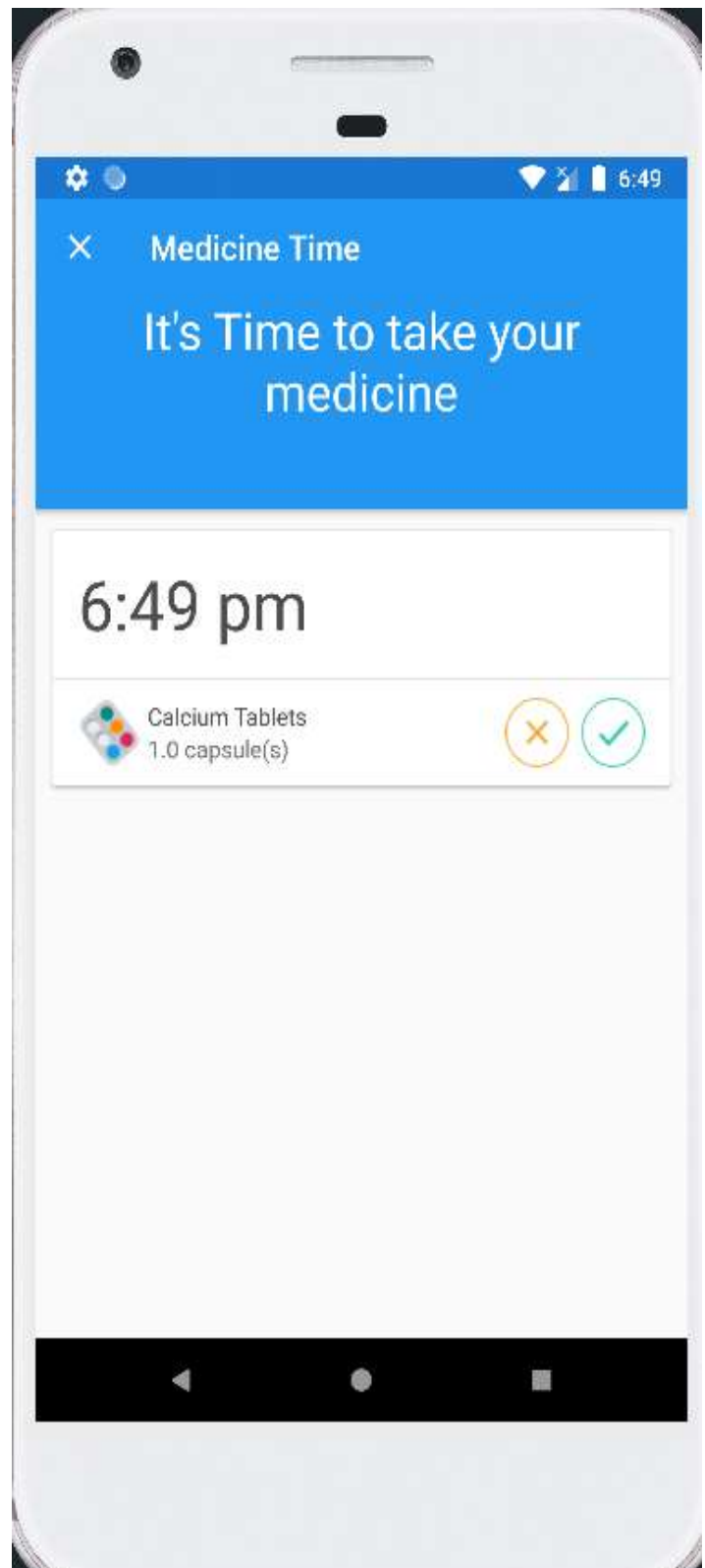


Fig 4.1.8: Notified via alarm



CONCLUSION

Thus by referring many existing products, previous projects and research papers based on medicine dispenser and also taking into consideration problems faced by disabled people, Geriatrics, etc. We thought of such a system which will help to overcome the disadvantage of existing or previous system. A mobile-phone-based automated medication reminder system shows promise in improving medication adherence and blood pressure in high-cardiovascular-risk individuals. The patients will get the schedule of medicine in-take time with medicine image, starting and automatic alarm ringing system and doctor's contact details. The scheduled reminder will not suggest any kind of medicine which is not prescribed by the doctor that will assure the safety of the patient and also will avoid wrong dosages. This will be done without any extra cost.

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